



Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning: A Cognitive Science Perspective

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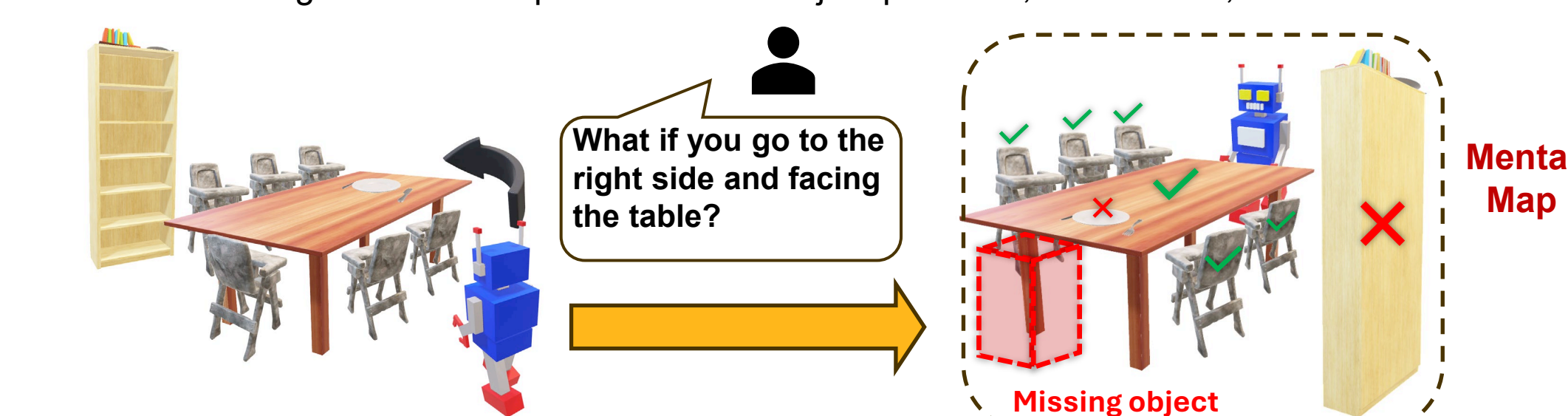
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Background and Motivation

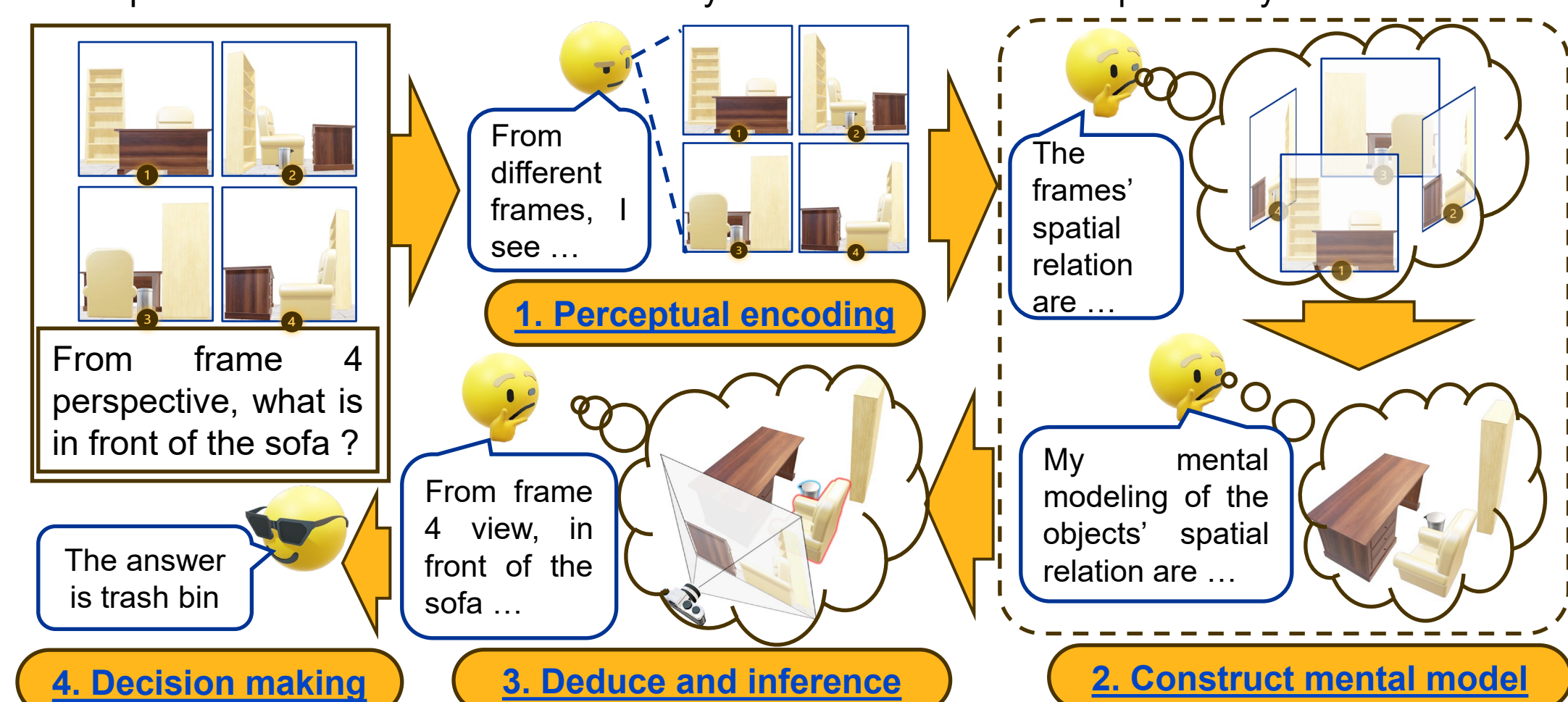
❖ Background: Multi-view Spatial Reasoning

- Multi-view spatial reasoning requires a model to integrate observations from different viewpoints while maintaining a coherent representation of object positions, orientations, and relations.

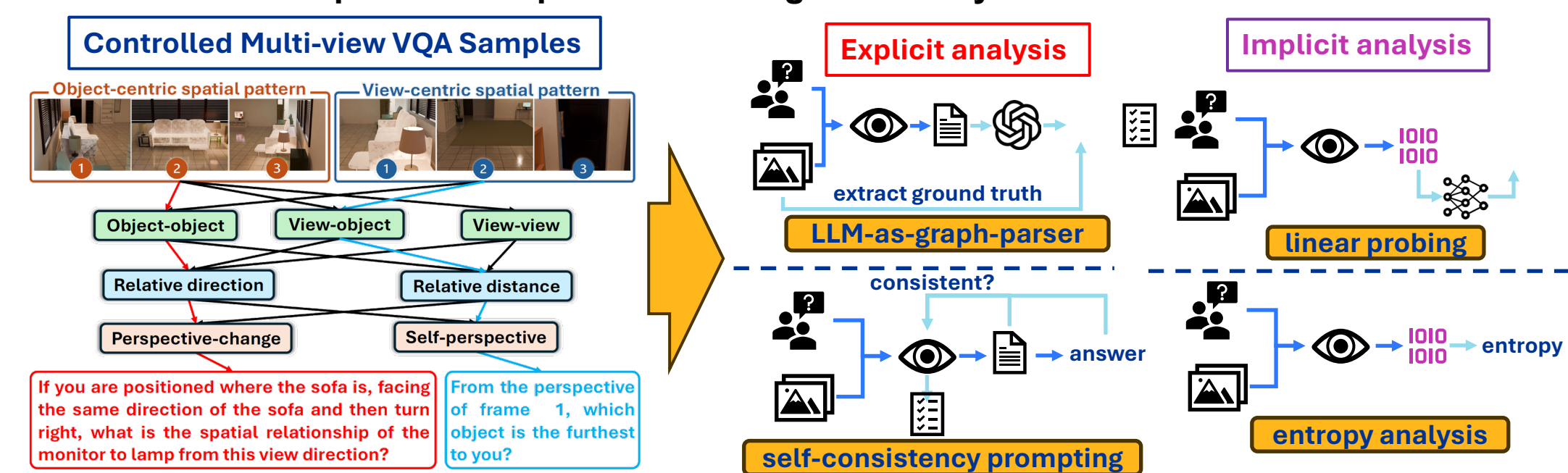


❖ Motivation: Spatial Reasoning from a Cognitive Science Perspective

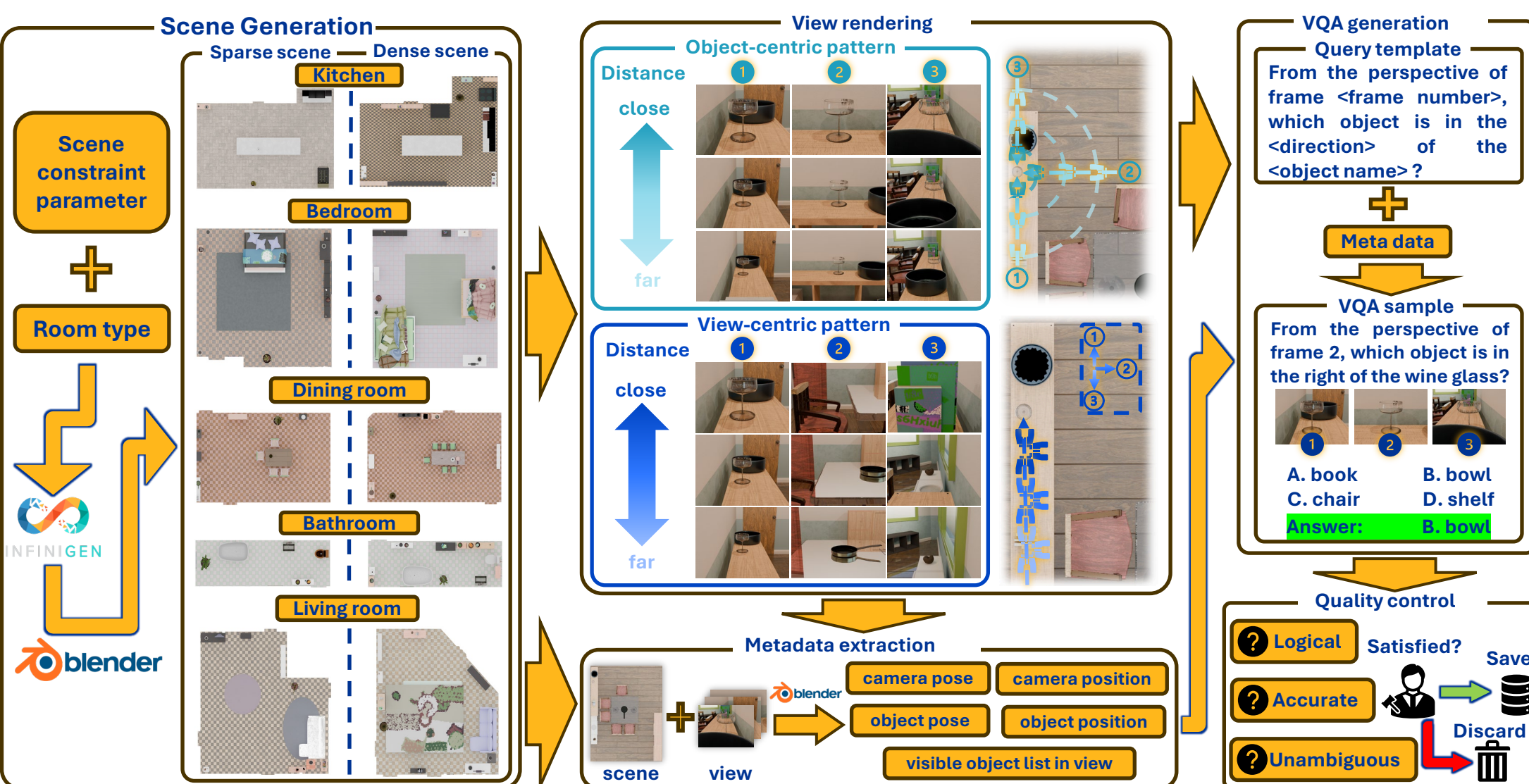
- Inspired by cognitive science, we analyze spatial reasoning as a sequence of four phases:
- This phase-based view allows us to identify failures that cannot be explained by final-answer acc.



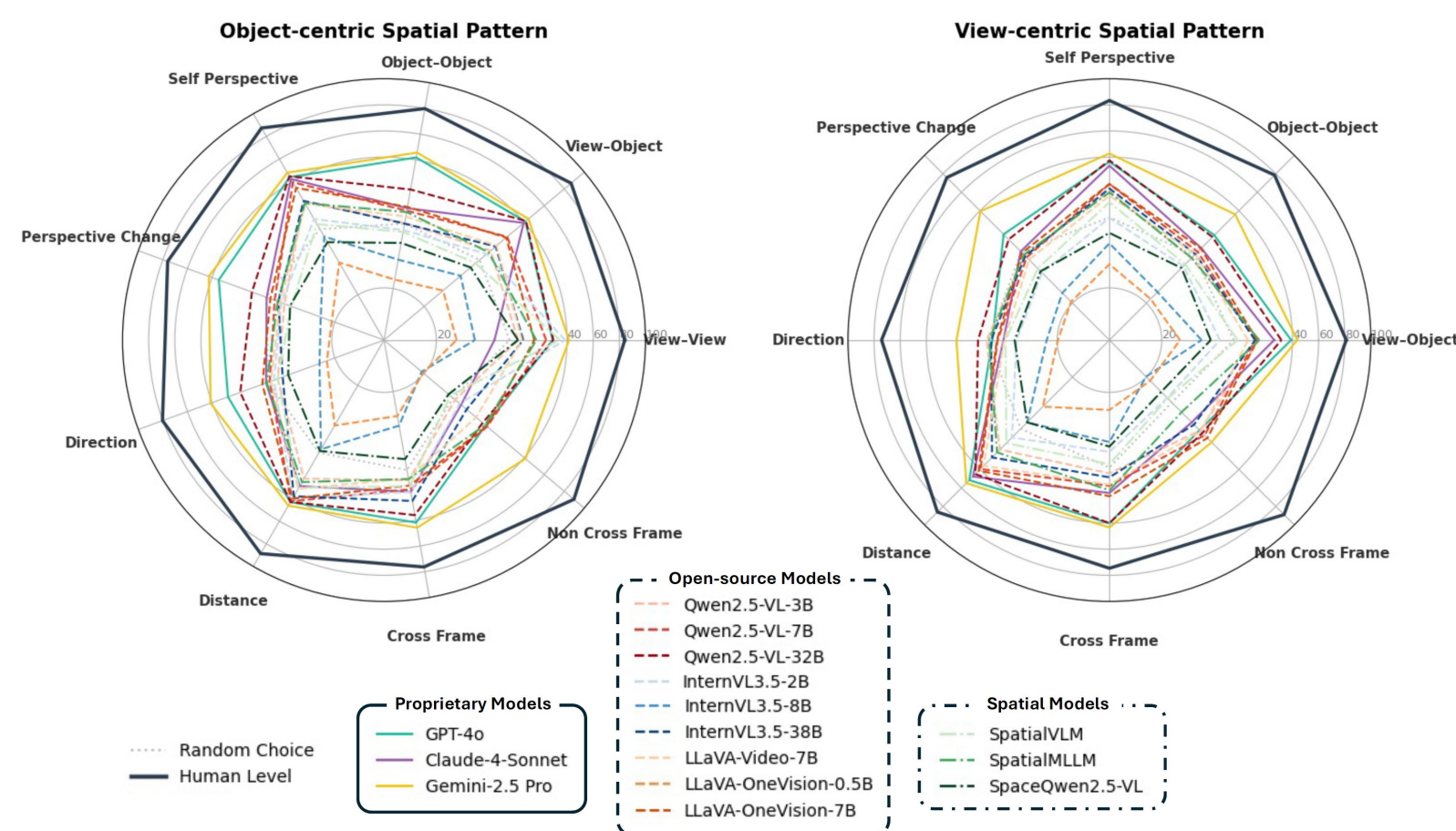
❖ Our Method: Explicit and Implicit Reasoning Path Analysis over Different Phases



Data Generation



❖ VLM Overall Performance



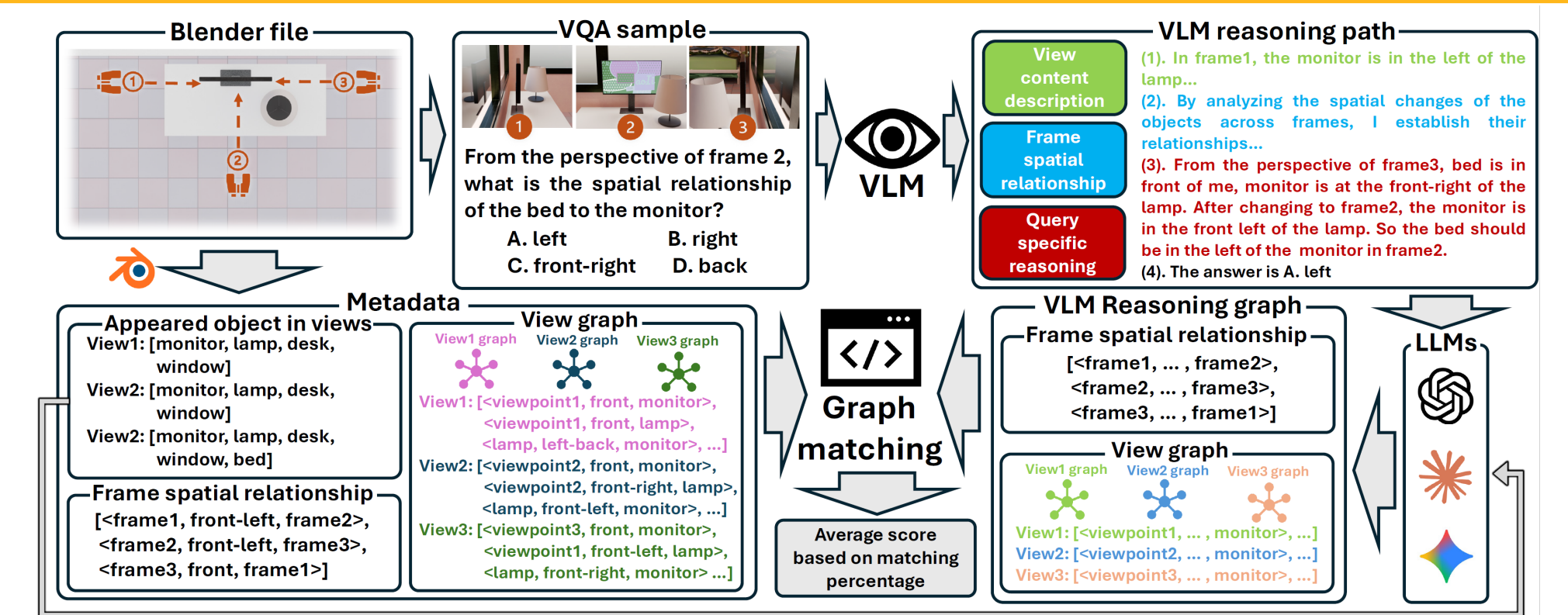
- Object-centric reasoning is consistently easier than view-centric reasoning.
- Perspective-changing queries cause a performance drop of more than four percentage points.
- Cross-frame reasoning is substantially harder than within-frame reasoning.

Explicit VLM Spatial Reasoning Analysis

❖ Analysis using LLM as Graph Parser

Method:

- Extract visible objects and spatial relations from Blender metadata to construct ground-truth view and frame graphs.
- Prompt the VLM through four reasoning phases, then use an LLM to parse each phase into spatial triplets: (subject, relation, object).
- Match predicted triplets with ground-truth graphs to measure phase-wise correctness and locate failures in perception, cross-view alignment, or inference.



Findings:

- Models perform well in perceptual encoding but deteriorate during cross-view alignment and inference.
- Larger models degrade more slowly, showing greater phase stability.

❖ Self Consistency Prompting

Method: Re-evaluate the multi-view input, reasoning path, and answer to measure reasoning-answer agreement.

Findings:

- Correct answers exhibit substantially higher self-consistency than incorrect answers.
- Errors are associated with contradictory reasoning and unstable relational grounding.

Model	Correct (%)			Incorrect (%)		
	1	2	3	1	2	3
Qwen2.5-VL-3B	73.2	43.8	32.6	68.7	31.4	18.5
Qwen2.5-VL-7B	76.1	58.2	36.3	72.5	43.6	21.9
Qwen2.5-VL-32B	81.9	63.5	38.8	76.8	44.2	24.6

Table 1: Phase-wise reasoning correctness of Qwen2.5-VL.

Model	Correct (%)	Incorrect (%)	Overall (%)
Qwen2.5-VL-3B	64.8	32.5	48.6
Qwen2.5-VL-7B	74.1	41.7	59.3
Qwen2.5-VL-32B	83.6	54.2	68.4

Table 2: Self-consistency prompting of Qwen2.5-VL family.

Implicit VLM Spatial Reasoning Analysis

❖ Linear Probing Analysis

- Train probes on token representations from each phase to predict the final-answer signal.
- Use probe loss to measure how much task-relevant information remains accessible.

Findings:

- Probe loss increases across phases, indicating progressive information degradation.
- Object-object relations are most stable, while view-view relations degrade the most.

❖ Entropy Dynamics

Method:

- Measure the mean and variance of token-level entropy at each reasoning phase.
- Compare uncertainty patterns between correct and incorrect reasoning paths.

Findings:

- Correct reasoning maintains lower entropy, while incorrect reasoning remains more uncertain.
- View-centric, perspective-changing, and cross-frame queries introduce uncertainty at early, middle, and late phases, respectively.

